

CODE SPORT



In this month's column, we feature a medley of computer science interview questions.

This is the beginning of a new year. So I would like to wish all our readers a great and happy year ahead. As is our usual practice, we will start the New Year by featuring a set of computer science interview questions.

1. Let us consider the problem of predicting house prices in Bengaluru. You have training data for only 1000 data points. Each data point has 50 features.
 - a. What are the machine learning models you may want to experiment with?
 - b. How would you address the problem of over-fitting the model to the limited training data?
 - c. Are you familiar with L1 and L2 regularisation? What is the difference between the two?
2. There are a number of online news services where consumers can read what they are interested in. Given a set of online news articles, you first need to identify their categories, which can be politics, sports, business, entertainment, etc. You are asked to build a text classification system for online news articles. How will you implement this? How will you implement a supervised learning solution? If you are told that you don't have access to labelled data, how will you devise an unsupervised learning method for this problem?
3. In the above question, we considered a static set of news articles for classification. Consider now that you have a streaming feed of online news articles, and you need to classify each one of them as it appears in the main feed and then decide whether to add it to the feed of each individual user, based on his or her interests. How will you modify the above application?
4. Let us consider a binary classification problem for which you are asked to label emails as either spam or non-spam. You are given a large enough training data set (100,000 emails) in which the emails are already classified as spam or non-spam. Using this, you are asked to train a binary classifier that can mark an incoming email as spam or non-spam. However, there is one problem with the training data that you have been given. Among the 100,000 training data emails, only 100 are labelled as spam. All other training data is instances from the non-spam category. What would be the problem if you train your model with such an unbalanced training data set? How will you avoid this issue?
5. You are given two unsorted lists of integers. You are asked to create a sorted list consisting of all the elements in the two lists. Can you write an algorithm for the same? What is the worst case running time of your code?
6. You are given a list of N nodes in a tree, with each node represented by an integer. For each node, you are also given the details of its parent node. You are not given the edges of the tree explicitly, but are told that it is a binary tree. You are asked to write code to perform an in-order traversal of the tree using this information. How will you solve this problem: (a) if you are told that you can perform pre-processing on the information and can use any amount of additional storage; and (b) you are told that you can use only a constant amount of additional storage.
7. Given a large text document, you are asked to create a summary of the same. There are two types of summarisation algorithms, namely, extractive and abstractive summarisation. In extractive summarisation, the summary will contain sentences extracted from the text. In abstractive summarisation, the summary will contain the gist of the original text, without including the exact sentences from the text. What kind of summarisation algorithm will you choose? Explain the rationale behind your choice. Now assume there is a real-time conversation that is occurring. How will you create a running summary of this?